METRIC
A-A-59665B
8 June 2023
SUPERSEDING
A-A-59665A
11 February 2013

COMMERCIAL ITEM DESCRIPTION

MORPHOLINE, TECHNICAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for morpholine to be used as a corrosion inhibitor in heating plants where boiler feed water is deaerated.

2. CLASSIFICATION. Morpholine shall be classified by classes and sizes specified. The selected class and size of the product to be supplied shall be specified in the acquisition order (see 7.5(b)).

2.1 Class.

Class 1 - Morpholine 99.0 percent
Class 2 - Morpholine 91.0 percent
Class 3 - Morpholine 40.0 percent

2.2 Size.

A – 19 liter (5-gallon) can
B – 208 liter (55-gallon) drum

3. SALIENT CHARACTERISTICS

3.1 Purity. Morpholine shall conform to the purity of table I when tested as specified in 3.3.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: DLA Aviation, VEB, 6090 Strathmore Rd, Richmond, VA 23237 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil.

AMSC N/A FSC 6810

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TABLE I. Purity requirements for morpholine.

Requirement	Class 1	Class 2	Class 3
Purity, weight %, minimum	99.0	91.0	40.0

3.2 Test method. Into a one-liter volumetric flask, place approximately 14 grams (g) of morpholine, weighed to the nearest 0.01 g. Dilute to volume with distilled water and mix thoroughly until homogeneous. Transfer a 25 milliliter (mL) aliquot to a 250 mL wide mouth Erlenmeyer flask and titrate with standardized 0.1 M hydrochloric acid against methyl red indicator. Calculate the purity of morpholine by:

$$\% \text{ morpholine} = (M \cdot V \cdot 348.4)/W, \text{ where}$$

M = molarity of hydrochloric acid
V = volume, mL, of hydrochloric acid
W = weight, g, of sample

3.3 Workmanship. Morpholine shall be a clear liquid, visibly free from foreign matter, and suitable for the intended use.

4. REGULATORY REQUIREMENTS

4.1 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations § 1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

4.2 Recovered materials. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance. Any certificate of analysis (COA) normally provided with the product to the commercial market shall be provided with each lot or batch of product. The lot or batch number and the date of inspection/analysis from the offer shall be on each COA.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

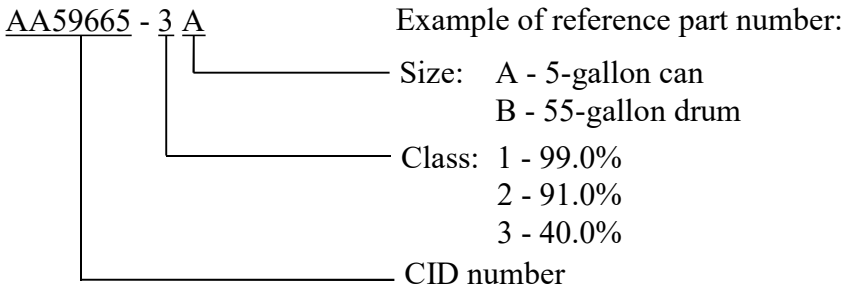
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6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.5(c)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



AA59665 - 3 A indicates: a solution that contains a minimum weight of 40 percent morpholine in a 5-gallon can.

7.2 Classes. Selection of class may depend upon environmental conditions of storage and handling. To aid selection of class, the approximate physical properties of each are provided in table II.

TABLE II. Physical properties of morpholine.

Property	Class 1	Class 2	Class 3
Specific gravity @ 20º /20 ºC	1.002	1.022	1.032
Freezing point, ºC	-3	-18	-27
Flash point (TOC)[1], ºC	38	46	100

[1] Tagliabue open cup.

7.3 Sources of documents.

7.3.1 Electronic copies of CFR documents may be obtained from https://www.ecfr.gov.

7.3.2 Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/browse/index/far.

7.3.3 Copies of FED-STD-313 may be obtained from https://quicksearch.dla.mil.

7.4 National stock number (NSN). The NSNs listed below are assigned for the listed product types and sizes (see table I). Other NSNs may also correspond with this document.

NSN	Class	Size
6810-00-419-4298	3	A

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6810-00-559-9889	3	B
6810-00-616-9437	2	B

7.5 Ordering data. Acquisition documents must specify the following information:

a. CID document number, revision, and CID PIN.
b. Product class (see 2.1) and size (see 2.2).
c. Packaging requirements (see 6.1).

7.6 Subject term (keyword) listing.

boiler
corrosion inhibitor
heating plant

MILITARY INTERESTS:

Custodians:
Navy – SH
DLA – GS

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS

Preparing activity:
DLA - GS3

(Project 6810-2023-004)

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